

In-Class Quiz: April 7

Points: 8/8

✓ **Correct** 1/1 Points

1

Suppose the mdecode microinstruction is accidentally omitted from the microcode preamble. What is the most immediate consequence of this error during the instruction cycle? 

- ☐ The program counter (pc) will not increment by 4.
- ☐ The instruction register (ir) will remain empty.
- ☒ The mswitch instruction will fail to route to the correct microcode routine.
- ☐ The shared bus will experience a fatal collision.

✓ **Correct** 1/1 Points

2

Look at the provided microcode for the mov instruction (register operand path). An optimization proposes replacing mb .rw with the two lines from .rw directly inline to save execution cycles. How many microinstruction cycles are saved per register mov instruction executed? 

```
mmove regSrc, rs2, <read>
mmove regData, regVal
mb .rw
...
.rw: mmove regSrc, rd, <write>
mb .begin
```

- ☐ 0
- ☒ 1

☐ 2

☐ 3

✓ **Correct** 1/1 Points

3

The microinstruction `madd reg1, imm, <args>` performs $\text{reg1} \leftarrow \text{reg1} + \text{imm}$ and sends `<args>` on the bus. In the standard preamble, `madd pc, 4` is used. What implicit action is deliberately ignored here? 

☒ Sending an arithmetic operation type to the ALU.

☐ Reading a value from the register file.

☐ Extending the sign of the immediate value.

☐ Checking the state of the flags register.

✓ **Correct** 1/1 Points

4

When implementing a conditional branch instruction like `bgt` in microcode, which microinstruction format **MUST** be utilized to alter the flow of the processor's execution based on the ALU flags? 

☐ `mloadIR`

☒ `mbeq`

☐ `madd`

☐ `mmov`

✓ **Correct** 1/1 Points

5

In the microprogrammed data path, the register file contains regSrc, regData, and regVal connections to the shared bus. If a designer attempts to execute `mmov A, regVal` and `mmov regSrc, rs1` in a single microinstruction cycle to save time, why will it fail? 

- ☐ regVal and regSrc share the same internal register file port.
- ☐ The ALU cannot accept an input into register A at the same time an address is provided.
- ☒ The single shared bus can only handle one data transfer at a time.
- ☐ rs1 is an external signal and must be routed through the ALU first.

✓ **Correct** 1/1 Points

6

A developer is implementing a custom memory store micro-operation. They have already correctly calculated the target memory address and placed it into the mar. To successfully write the data to memory, into which internal register must the data be moved before triggering the memory write operation via the shared bus? 

- ☐ regData
- ☐ ldResult
- ☒ mdr
- ☐ aluResult

✓ **Correct** 1/1 Points

7

A custom conditional branch instruction, "Branch if Less Than or Equal" (BLE), is being implemented in microcode. The ALU has just executed a compare (`cmp`) operation. Based on the available internal flag registers, which microinstruction correctly evaluates the condition to take the BLE branch? 

- ☐ mbeq flags.E, 1, .take_branch
- ☐ mbeq flags.LT, 1, .take_branch
- ☐ mbeq aluResult, 0, .take_branch
- ☒ mbeq flags.GT, 0, .take_branch

✓ **Correct** 1/1 Points

8

During the execution of any standard instruction, the program counter (pc) is automatically incremented by 4 during the microcode preamble. If the current instruction is a *taken* branch, how does the microprogrammed datapath resolve the final pc value to ensure the jump occurs? 

- ☐ The hardware preamble conditionally skips the madd pc, 4 instruction if a branch opcode is detected.
- ☐ The ALU subtracts 4 from the pc before calculating and adding the branch offset.
- ☐ The mswitch operation directly loads the branch offset into the pc instead of the microprogram counter.
- ☒ The specific microcode routine for the branch instruction overwrites the pc with the pre-calculated branchTarget.



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